

Data Analytics and Decision-making in Retail and Fast-Moving Consumer Goods

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Introduction

The most accurate definition of data analytics in modern business is using structured or unstructured data and formal, statistical, or machine learning analysis to derive insights that improve corporate decision-making. The result of modern-day data analytics gives the probabilistic result and helps to generate probabilities for various possible outcomes rather than firmly suggesting one deterministic result (Agrawal, 2014). With the advancement of internet technology, a large amount of data flow can be identified in the contemporary organization. Various sources and technologies such as personal computers, mobile phones, government records, healthcare records, and social media are the constant creators of the data; Thus, data are getting more extensive and complicated. In today's era, data analytics is used for decision-making in banking and security, social media and entertainment, communication, healthcare, education, retail and wholesale trade, and transportation. (Le & Liaw, 2017).

In recent times, statistical data have been used for decision-making in different areas of the business. A considerable amount of data is gathered by companies every day. However, these raw data cannot be used directly, but data being analysed by data measurement levels such as nominal, ordinal, interval, and ratio (Black, 2023). The valid and good-quality dataset is the lifeblood of business data analytics because, based on the data, management can make various decisions for the business's current and future (Albright & Winston, 2014). Four major types of data analytics help in the decision-making process, which are discussed below (Sarker, 2021).

A) Descriptive Analysis:

It is the analysis of historical data and represents the changes that occurred in the business. Managers can identify the firm's strengths and weaknesses from past data.

B) Diagnostic Analysis:

The purpose of diagnostic analytics is to identify the root cause of the problem. The manager can compare coexisting trends or movements and establish causal relations between variables.

C) Predictive Analysis:

Many organisations utilise predictive analytics for risk assessment in the workplace, spot future market trends, and advance their company operations by comparing historical data with current industry trends.

D) Prescriptive Analysis:

This analysis is specifically used when an organisation makes data-driven decisions, as analysis helps managers decide what action should be taken.

This report aims to analyse data-driven decision-making to solve contemporary business problems. For this purpose, distinctive scholarly journals and peer-reviewed articles from the Robertson Library are selected to support the arguments. Furthermore, this paper discusses three current business issues in retail and Fast-Moving Consumer Goods (FMCG) because the retail industry, specifically FMCG products, is significantly less profit margin business. Thus, it is worth getting insight into how data analytics help solves retailer issues regarding Advertisement and promotion, Inventory Management, and Pricing and discounting strategy. Compared to the past several years, the retail industry has changed from unordered and traditional stores to elegantly designed and appropriately managed stores, supermarkets, and

shopping malls with outstanding ambiance and comfort. However, modern retailers' main issue is consumers' concern about brand, quality, price, and product availability (Verma et al., 2020).

Advertisement and communication management

Effective communication in consumer-packaged goods helps to enhance the sales and successful launch of the product in the market (Munir Dad, 2012). However, according to Munir Dad, (2012), FMCG products are relatively low prices and produced in large quantities to maximize profits through economies of scale. Pongiannan, (2014) described as FMCG products require a few thoughts before making a purchase decision. Munir Dad, (2012) further supports this argument and says that consumers spend little time making decisions before buying as these are low consumer involvement products. Thus, in the cutthroat competitive market, the response of advertisement and promotion management on customer buying behaviour and sale can be addressed through data analytics.

An appropriate advertisement campaign can create positive customer response and increase the sales of FMCG products despite the low-profit margin and customer involvement product. As per Pongiannan, (2014), low involvement products (FMCG) advertisements influence the customer's buying behaviour and increase consumer packaged goods sales. Pongiannan, (2014) has analysed various variables such as quality, frequency, medium of the advertisement, age, and gender and used data analytics techniques such as frequency analysis, Chi-Square test, t-test, Kruskal Wallis test, and descriptive statistics. Similarly, there is a positive correlation identified between advertisement expenses and the sales of the products. As expenses increase, sales also increase (Sen, 2021). The researcher has identified advertisement and promotional expenses as independent variables and gross sales as dependent variables. Descriptive statistics of multivariate regression analysis showed that as advertisement expenses increased, the product's gross sales increased (Sen, 2021).

Data analytics about the quality, frequency, and medium of ads and customer age and gender data helps management make decisions about marketing new products and allocating a budget (Pongiannan, 2014). Furthermore, data analysis can help the manager identify which product is needed for more advertisement because FMCG product ads require a high level of creativity to generate a response among the customer (Pongiannan, 2014). Additionally, Sen, (2021) emphasised that with the help of data analytics, companies can analyse the different advertisement materials for effective promotion, such as branding, social media marketing, and celebrity endorsement. The manager can decide on the advertisement's budget based on data analytics. Also, data analytics helps the manager identify the various products that require more advertising strategies in the rural area to reach that customer base (Sen, 2021).

Supply chain and inventory management in the retail sector

The key critical competitive component in the retail or FMCG sector is inventory management which is directly linked with customer service and relationship management. In the retail sector, the non-availability of the product led to poor customer service, and competitors took great advantage to push their products to the potential customer (Salam et al., 2016). Nemtajela & Mbohwa, (2017) also say that the firm's success depends on proper inventory management as it is the balanced relationship between minimising the inventory cost and maintaining good customer relationships and considering the uncertainty of the demand for such products. On the other hand, retailers can store more inventory and keep their customer services high; however, maintaining a high list leads to extra costs for the company. In contemporary business

management, data analytics helps the manager make appropriate inventory and supply chain decisions in the retail business context (Salam et al., 2016). Thus, there is a significant question about the appropriate decision, whether inventory should be maximized or minimised for retail businesses.

Proper inventory management and smooth supply chain work will increase profitability and customer service, specifically in fast-moving consumer products. Salam et al. (2016) say inventory level positively impacts customer satisfaction and buying behaviour. In the same vein, based on the structural equation modelling analysis technique, inventory management practices adopted by the firm positively impact the organization's performance and customer service level. Salam et al., (2016) have analysed various variables such as product dispatched information, lead (transit time) for each development, replenishment cycle, average sales, and inventory on hand. In contrast, Kamilah analysed different variables in the inventory management practices and knowledge, such as several employees, average sales of the firm, business establishment (duration), education and economic order of quantity, and type of the business. Salam et al., (2016) have analysed by simulation that reducing the inventory holding cost is an efficient and effective decision for the retailer. Moreover, Kamilah Ahmad, (2018) arguably said that the accuracy of the information provided by the supplier and the information received from the analytics of customer data play an essential role in the success of inventory management.

Data analytics helps managers in different ways while managing the stock. Quantitative data analytics helps maintain effective transactions with suppliers and customers, which allow the manager to make decisions related to cutting down the non-value-adding product inventory from the existing supply chain (Salam et al., 2016). Additionally, data analytics is an essential tool that enables managers to analyse the details of the various suppliers, such as price, quality, quantity, and supplying capacity. It will lead to proper inventory maintenance and enable managers to make appropriate decisions (Kamilah Ahmad, 2018). Data analytics and descriptive statistics are implemented to maintain raw material or finished goods inventories at levels that deliver the highest customer service levels at the lowest inventory maintenance cost. If suppliers fail to fulfil orders, delivery times lengthen, sales decline, and customer service decline. Management can save inventory costs and enhance decision-making along the whole supply chain by exchanging information such as demand, sales orders, and inventory status utilising collaborative information-sharing methodologies, including collaborative planning, forecasting, and stock replenishment (Salam et al., 2016).

Effective pricing and discounting strategy in retail

In recent times with the increase in the use of technology, a significant boom has been observed in the e-commerce of fast-moving consumer goods. For online and offline purchases of consumer goods, customers analyse various aspects such as price, brand, product quality and any discount given on the purchased product (Noor, 2020). Competition in pricing and discount has a beneficial impact on the customer buying behaviour and purchase intention. According to Noor, (2020), customers are more likely to buy more unplanned products or buy the product suddenly, which is termed impulse buying behaviour. In consumer goods pricing, sellers need to focus more on deciding the reasonable price of the product because of competition.

Appropriate pricing and discounting strategy increase the sales and profitability of the firm. According to Noor, (2020), pricing discounts and store displays play a vital role in increasing

impulse buying. Similarly, Luo et al., (2021) agreed that the deal period also plays a crucial role in increasing impulse buying in e-commerce marketing. However, Allender & Richards, (2012), arguably said that discounting and different pricing techniques do not affect unplanned buying (impulse buying) among the loyal brand as they are more likely to buy products from the specific brand without considering the price. Luo et al., (2021) analysed distinctive variables such as type of pricing discount (value-based price or psychological price), anticipated regret (regret for not buying the product which is on value), quantity-limited discount, time-limited discount and performed regression analysis, reliability analysis, validity analysis, and variance analysis.

Data analytics on the customer purchase records and payment records helps the manager decide appropriate pricing strategies to increase impulse buying, which leads to high sales (Noor, 2020). Data analytics can address consumers' varying impressions of various product categories; thus, retailers should modify their pricing strategy and price discounting time per the types of marketed items (Luo et al., 2021). Furthermore, data analytics is crucial as it gives insight to the retailer about which products need to be discounted and which brand should be put for pricing promotion and generate more profit (Luo et al., 2021).

Conclusion

This report is designed to critically analyse the contemporary business issues in the retail industry context and data-driven decision-making in different areas in retail and FMCG contexts. Advertisement and communication, inventory management, and pricing strategy are three contexts discussed, and a literature review has been done. According to the researcher (Munir Dad, 2012; Pongiannan, 2014), there is a positive relationship between product advertisement and sales of consumer packaged goods even if the low consumer involvement product and low-profit margin. Moreover, this report has analysed how inventory management and proper supply chain affect customer satisfaction and loyalty. Researcher (Kamilah Ahmad, 2018; Salam et al., 2016) examined different variables such as dispatched information, the transit time of the products, education, type of business, and economic order of quantity and found that reducing the inventory cost benefit the organization as well as it is also essential to maintain good relation with supplier. On the other hand, pricing strategies and discounting play a vital role in increasing customer impulse buying (Luo, 2022; Noor, 2020).

Data analytics is used in various business areas to make data-driven decisions in the retail and FMCG products context. Management can decide on advertisement and promotion management, product discounting, inventory costing, customer satisfaction, and brand loyalty. Besides this, data analysts should properly know data analytics types and data measurement levels to apply various analytical tests. Overall, it can be summarized that data analytics and data-driven decision-making helps retailer and other business to make a better decision to improve operations, sales, profit, inventory, customer satisfaction, and brand loyalty.